PROJECT PLAN / 01

FutureProof

Housing growth, transit readiness and research with planners

Plan the service before the homes open.

FutureProof is a Toronto corridor-planning prototype. It connects real housing proposals to a passenger-and-bus simulation, compares affordable transit packages across occupancy phases, and helps a planner investigate why a plan fails or which assumption deserves another look.

The core decision
For the same planned homes, what transit service needs to be ready at move-in, what resources does it require, and how sensitive is the answer to demand and delivery timing?

Two connected parts

1. Planning workspace. Inspect developments, edit explicit demand assumptions, compare existing service, a priority lane, additional scheduled service, and their combination. See stop-level consequences and export an evidence-linked brief.

2. Papiers research partner. Read the team's research context, notice patterns in computed results, propose a hypothesis, run a controlled simulation experiment, evaluate it, and select a next investigation. Human knowledge can redirect that sequence.

User story

A transportation planner is reviewing three developments opening in two phases along one TTC bus corridor. The planner loads the sites, checks occupancy assumptions, and compares service packages under separate capital, operating and fleet limits. A downstream stop still performs poorly. The research partner investigates the bottleneck using a question or annotation from Papiers; the planner challenges an assumption, and the agent changes its next experiment. The final brief records the package tradeoffs, findings, open questions and limitations.

Benefit and intended distinction

The planner gets one inspectable chain from housing assumptions to service outcomes to follow-up research. Residents and community groups can read the same comparison. The intended distinction is the combination of phased housing readiness, downstream passenger effects and a research history shaped by human knowledge. Transport modelling itself has established precedent.

Hack the Valley 11 | Planning version: October 16, 2026. This document specifies intended behavior; implementation, access and real-world validity remain to be demonstrated. Flags F1-F7 carry concerns from the review: amber marks open items; blue marks plan decisions.

PROJECT PLAN / 02

Scope and product features

One complete planning-and-research workflow

Area	Committed target
Study area	One direction of one TTC bus corridor; approximately 8-12 stops and three reviewed developments.
Time and demand	Two occupancy phases; one selected peak arrival window; low, central and high demand; one delayed-upgrade case.
Packages	Existing service, a reviewed priority-lane option, an additional-service timetable, and both. Housing stays fixed between package comparisons.
Research	One completed, evaluated agent-led investigation; an agent-generated next step; one visible human redirection; genuine Papiers use in at least one direction.
Outputs	Map/list view, comparison cards, simple vehicle/queue replay, research history, and a saved planning brief with assumptions and evidence.

Essential interactions

Explore. Select a building or stop; inspect its source, planning status, assumed occupancy, walking access and demand contribution. Distinguish actual proposals from simplified visual masses.

Compare. Edit occupancy, demand assumptions, budgets and intervention dates. Show current conditions, growth with existing service, and growth with each feasible package. Mark old results stale after edits.

Investigate. Ask why a stop worsens, review the agent's hypothesis and experiment, inspect evidence, add judgment, and see the next research choice change.

Understand and save. Provide source/assumption labels, keyboard controls, readable text alternatives, a 2D/list view and pause for motion. Save the scenario, results and research trail together.

OPEN F2 | Protect the complete workflow
These targets remain conditional on the first data and runtime checks. Cut detailed 3D context, replay scrubbing, extra chart polish and redundant AI chat before cutting the simulator, comparison or Papiers research loop. If both phases cannot be completed, explicitly disclose a one-phase fallback rather than presenting it as the full target.

Outside this build

Signal priority is removed entirely. Also excluded: arbitrary building placement, city-wide coverage, transfers, mode-choice or emissions prediction, heat modelling, new vehicle types, automatic code modification, and multi-agent orchestration. The research agent uses the same bounded simulator and existing inputs; it does not require another urban model.

PROJECT PLAN / 03

Data and demand

Real geography; explicit assumptions where observations are missing

Input	Use and readiness check
Housing [3]	Select three real proposals; retain IDs, units, coordinates, status and record date. Use net added units for growth and check related applications/replacements for double counting.
TTC GTFS [4]	Load stops, ordered trips and shapes for one service date and direction. Resolve branches and calendar exceptions; freeze a local input bundle.
Baseline riders [5]	Seek stop/time observations and incoming loads. Route totals are only a coarse check; they do not establish peak stop-level demand.
Walking and buildings [6,7]	Prepare a small, reviewed pedestrian-access table and optional clipped massing. Check barriers and crossings; retain attribution and data vintage.
Packages [8]	Specify affected links, travel-time effects, added dispatches, full-cycle fleet needs, costs, availability phases and effect ranges.

Demand construction

One visible calculation chain
Net added units x occupied share x residents per occupied unit x trips per resident in the peak window x transit share. Then allocate those trips to accessible stops, times, direction and destinations.

Every factor needs a unit, denominator, source or assumption label and allowed range. Keep existing riders separate from added residents. Generate consistent passenger origins and destinations. Incoming buses may already be occupied; riders travelling beyond the corridor remain accounted for at its boundary.

OPEN F1 | Choose the corridor and baseline
Data owner closes this before full modelling: name the route/direction, service date, stop IDs, three developments and available files. Record a low/central/high background-demand and incoming-load profile if observations are missing. In that case, describe the build as a scenario experiment, not a calibrated TTC forecast.

OPEN F5 | Fill the numerical intervention catalogue
Data and simulation owners must agree on one lane scheme and one extra-service timetable, with effect ranges, capital cost, annual operating cost, price year, fleet increment and availability. Unknown values remain labelled assumptions. A paper can motivate an experiment without supplying a corridor-specific effect size.

Preserve evidence

Save the exact prepared input bundle, source links, retrieval dates, licences and content hashes. Version occupancy dates as scenario assumptions unless evidenced. Low/high cases vary named uncertain factors; they are not assigned real-world probabilities. No live feed or city-wide download is required during the demo.

PROJECT PLAN / 04

Simulation and decision rules

Consistent passengers, fair comparisons and visible tradeoffs

Use a small discrete-event model: passenger arrival, bus arrival, alighting, capacity-limited boarding, dwell and departure. Queues persist when buses are full. Boarding-related dwell feeds into later bus spacing. Define one stop-berth rule and whether overtaking is allowed. A lane changes link travel-time/delay assumptions; extra service changes the dispatch timetable and resource requirements.

Outcome	Report
Passenger service	Mean and 95th-percentile wait; unique share denied boarding at least once; remaining queue and unresolved passengers, by stop and rider cohort.
Journey and reliability	Within-segment ride time and wait-plus-ride time for matched origin/destination cohorts; headway variation. Incoming onboard riders have separate segment ride-time results.
Resources	Capital cost, annual operating cost, full-cycle fleet requirement and phase availability, shown beside outcomes.

F3 | Addressed in this plan: measure the lane's travel benefit
Add in-vehicle and within-segment journey time. The lane can help passengers even if scheduled frequency stays constant. These measures exclude walking, transfers and travel beyond the modelled segment.

Readiness and package selection

The user sets a P95-wait limit and a denied-boarding limit. A package passes only if every evaluated stop meets both limits in every selected demand case for that phase, and capital, operating, fleet and availability constraints all hold. Show aggregate passenger metrics across repeated runs plus variation between runs; a percentile is not a confidence interval.

Return the passing packages and their cost/service tradeoffs. Remove dominated options from the preferred shortlist but keep them inspectable. If one recommendation is requested, apply an explicit user-selected priority, such as lowest annual operating cost among passing packages, then capital cost as a tie-breaker. Never add one-time and recurring costs into an unexplained total.

OPEN F4 | Set the actual acceptance values
The rule above is the proposed default. The team must select demo thresholds and a cost priority before the first comparison. Label them user-selected, not official TTC standards. Sparse or unresolved wait observations cannot establish a pass; show an indeterminate result or failure with the outstanding queue. If nothing passes, report that outcome without reducing the homes.

Fairness and boundary checks

Match housing, passenger journeys and pre-generated random draws across packages. Warm up the model, measure a defined arrival cohort, then allow later service to observe its waits. At an analysis cap, retain unserved riders and wait lower bounds. Faster lanes never create free trips; fleet and service costs use a full round trip including return and layover.

PROJECT PLAN / 05

Papiers: the research partner

Investigate what the planner may have overlooked

Make this a central product capability and a target sponsor submission. It extends the planning comparison into a research process: the agent uses human knowledge to decide what to test, executes the test, evaluates evidence, and chooses what to investigate next. The supplied challenge requires at least one completed investigation, its evaluation and an agent-generated next step, with Papiers used in at least one direction. [1,2]

Human context Papiers + feedback → **Hypothesis** bounded question → **Experiment** validated simulator → **Evaluation** metrics + evidence → **Next step** agent chooses

Repeat within a fixed experiment budget; request human judgment when evidence is missing.

Stage	Concrete behavior and saved evidence
Read context	Retrieve a team-selected Papiers paper, question or annotation through supported access. Save source IDs, relevant excerpts and which human question informed the investigation.
Notice and hypothesize	Inspect computed queues, denials, headways and phase comparisons. Propose a bounded explanation, expected observable effect, controlled change and evaluation metric.
Run an experiment	Validate parameters and resource limits; freeze housing and other controls; execute paired runs through the same simulation worker. Save configuration, seeds and run IDs.
Evaluate and continue	Use computed differences and run variation to label the hypothesis supported, contradicted or inconclusive within the model. Choose a new question from the evidence, or request human expertise.
Teach back	Present the finding, failed tests, caveats and next question. Accept a human correction and record how it changes the next experiment. Write a finding to Papiers if supported.

A small experiment vocabulary

Policy tests use only the existing lane, service and combined packages, plus their availability dates. Diagnostic tests vary uncertain incoming loads, departure-time concentration, boarding/alighting dwell parameters or lane-delay effects within declared ranges. Diagnostic changes reveal model sensitivity; they are not free or automatically actionable interventions.

F6 | Coherence decision
Papiers supplies the human research context for the experiment loop. The agent tests existing service packages and declared uncertainty ranges. A single agent with a few validated tools is enough. Its completed experiment, evaluation and next step demonstrate the research contribution.

The agent can surface overlooked consequences represented by the model, such as a downstream queue hidden by the corridor average. For an unmodelled cause, it opens an evidence request instead of claiming to discover a real-world fact.

PROJECT PLAN / 06

A research demonstration

An illustrative investigation, not a prewritten finding

Starting observation

The planner compares the same housing phase across the four packages. A downstream stop has frequent denied boarding, although total corridor demand looks manageable. A Papiers note questions the assumed load of buses entering the segment.

Step	What the live system must do
Hypothesis	The agent proposes that incoming occupancy, rather than only new housing demand, drives the downstream bottleneck. It states a measurable prediction and cites the motivating note.
Controlled test	Run lower and higher incoming-load cases with the same housing, dispatch plan and other assumptions; use matched seeds. Treat this as an uncertainty investigation, not a policy improvement.
Evaluation	Compare stop-level denial share, P95 waits, queues and journey time. Show actual values and run variation. If the evidence is weak or contrary, say so.
Human redirection	The planner contributes a constraint: no extra vehicles are available in phase one. The agent records this and stops proposing infeasible phase-one frequency increases.
Agent-generated next step	The agent chooses a relevant remaining test from the results and updated constraint: for example, lane-effect sensitivity or upgrade delay. Its choice must respond to evidence, not replay this example verbatim.

Papiers integration contract

Minimum: a real supported read retrieves human context that changes a hypothesis or experiment. **Preferred:** also write a concise finding back with source IDs, run links, assumptions, limitations and next question. One direction satisfies the supplied integration requirement; both directions strengthen the feedback story. [2]

OPEN F6 | Access is the first research gate
Research owner must test the sponsor-provided MCP/API or another sponsor-approved integration path immediately. The supplied guide is marked Coming soon; authentication, available fields and read/write support are untested. Demonstrate one real operation before building against assumed endpoints. A mock or manually copied note may support development, but is not presented as live Papiers integration.

A bounded autonomous loop

A user starts an investigation. The agent may run up to three experiment batches within a wall-time and API-cost budget chosen at kickoff. It can stop, request expertise or mark a result inconclusive. Save human feedback as a versioned research input; lock physical and budget constraints. It cannot change simulator code, invent observations or convert a diagnostic assumption into a recommended policy.

Completion target: one experiment and evaluation, a next step chosen by the agent, and visible human redirection. Execute the follow-up too if runtime permits. A negative or inconclusive result still demonstrates useful research.

PROJECT PLAN / 07

Architecture and runtime

One simulation engine serves planning and research

Component	Responsibility
React + TypeScript	Scenario forms, MapLibre corridor view, accessible list, comparisons, simple replay and research history. Building extrusion is optional polish.
FastAPI + worker	Validate scenarios and experiments; queue CPU-heavy simulation jobs; expose status, results and export. A job table and polling are enough.
Python model + prep	Prepare local spatial/GTFS inputs; run passenger and vehicle events headlessly; compute metrics and conservation diagnostics.
PostgreSQL / Tiger Data	Version sources, scenarios, runs and investigations. Use Tiger Data only if actual access works; PostGIS is conditional on verified support and a useful spatial query.
Gemini + Papiers adapter	Gemini proposes structured hypotheses, tool requests and grounded explanations. The adapter maps verified Papiers operations into the research workflow; set its contract after the access test. [9]

Shared contract before splitting work

Freeze one sample scenario, result and experiment record: site/stop/link IDs, coordinates, units, phase, demand profile, intervention IDs, budgets, seed configuration, metric definitions and status. A hypothesis record includes its source context, controlled changes, evaluation rule, run IDs, finding, human feedback and next step.

The frontend, replay, export and AI explanations all consume the same backend results. Every displayed number traces to a run. Save exact prepared inputs, model version and random draws; cache by the complete validated configuration. Retain aggregate metrics for all runs and detailed events for selected replays. Keys stay on the backend.

OPEN F2 | Benchmark before promising instant results
Engineering targets, not measured claims: a selected-phase central-demand four-package preview within 10 seconds; a full two-phase, three-case, 20-run matrix within 60 seconds on the demo host. That matrix is 480 runs before baselines, delay tests or research. Measure it early, show jobs asynchronously and label previews with run count. Never use a preview alone to certify readiness.

Reduce work without changing the evidence

Compute only the selected phase/case for exploratory previews; cache identical results; generate detailed replay events for one run. The research agent reuses the same worker and cache. Retain the full declared comparison for the final brief. If Monte Carlo variation is unstable, add runs or report uncertainty rather than implying that 20 runs is sufficient.

If database or hosting setup stalls, use a simple local persistence fallback and disclose which sponsor integrations are absent. If Papiers access fails, keep the planning product usable while resolving access; do not claim that its challenge has been demonstrated.

PROJECT PLAN / 08

Build plan and completion gates

Integrate early; name the owners at kickoff

Owner slot	Primary deliverable and coordination
A / Interface	Forms, map/list, comparisons, replay and research panel. Help D with integration and own deployed-flow verification.
B / Data + demand	Reviewed corridor bundle, assumptions and package inputs. Pair with C on baseline and boundary conditions.
C / Simulation	Events, metrics, invariants and experiment runner. Own hand-checkable tests and runtime measurement.
D / Research + API	Papiers access, agent loop, API and persistence. A shares integration/export so this role does not absorb every backend task.

Gate	Evidence required before expanding scope
First 2-3 hours	B names the corridor and files; C runs a tiny correct model; D proves Papiers/Gemini access; A renders the shared sample. Assign actual names and confirm deadlines.
Basic loop	A real data bundle flows through the API to current/growth results in the interface. Conservation and overloaded-queue tests pass.
Comparison + research	Four packages enforce resources; stop and journey metrics agree. One real Papiers-informed experiment completes, is evaluated, and generates a next step.
Robustness + demo	Complete declared phases/cases, delay test, uncertainty, human redirection, saved brief and simple replay. Benchmark on the actual host.
Final reserve	Freeze features at least four hours before the team's working deadline. Verify after reload, rehearse, submit repository link and confirm selected categories.

OPEN F7 | Named ownership and conflicting times
Assign A-D, one integration lead and one submission owner. Official Devpost pages conflict: the overview deadline is Oct 18 at 8:00 AM EDT; rules say 10:00 AM ET, with an initial submission Oct 17 at 9:00 PM ET. Supplied notes also differ on the start time. Confirm with organizers; plan against the earlier final time until resolved. [1]

Verification that earns completion

Check passenger conservation, capacity, ordered events, persistent denied riders and a deliberate overload. Verify reproducibility, phase/fleet/budget rejection, matched comparison cohorts and unresolved-wait handling. Check that chart, replay, export and AI numbers agree. Test one human correction changing the agent's next configuration; audit the actual Papiers operation. Have a new user explain the selected package and its limitations.

If late: remove visual polish and scrubbing first, then extra research cycles and optional writeback. Preserve a genuine Papiers read, one completed evaluation and an agent-generated next step. Never relabel an untested integration or an incomplete comparison as finished.

PROJECT PLAN / 09

Value, differentiation and fit

Compete on a clear decision and a visible research process

Existing approach	Overlap and our proposed emphasis
CityCast [10]	Already combines development, household and transit changes with transport simulation. FutureProof emphasizes move-in phases, affordable service packages and an inspectable follow-up research trail.
UrbanSim [11]	Already connects development timing, land use and transportation scenarios. Our narrower product focuses on the service available when specific housing becomes occupied.
Replan [12]	Already combines transit planning, costs, vehicles and simulated demand. Our proposed emphasis is housing readiness plus Papiers-informed investigations and visible human redirection.

These are positioning judgments based on public descriptions, not proof that competitors lack equivalent features. The project's technical contribution is the working, reproducible integration. Validate the workflow with a planner before claiming that it improves professional decisions.

Tracks and themes

Target	Fit and what the demo must prove
School of Cities / primary	A concrete Toronto civic decision using real development locations and service data. Show usefulness, urban insight, accessibility and responsible interpretation. [1]
Papiers / primary	Actual autonomous investigation, evaluation and next step; real Papiers use; a human note or constraint visibly changes the research trajectory. Strong conceptual fit, conditional on execution and access. [1,2]
AI for Social Impact	Help civic users ask questions, understand tradeoffs and challenge assumptions. Demonstrate grounded answers and a useful task, without claiming measured policy impact.
Climate Action	Investigate transit service that can support housing growth. Show waits, reliability, travel time and resources. This model does not establish avoided car trips or carbon savings.
Gemini / secondary	Use real API calls for structured experiment planning and result explanations, validated by the backend. A generic summary alone is a weaker demonstration. [1,9]
Tiger Data / Vultr	Enter only if actually used: inspectable saved runs/queries for Tiger Data; a working hosted API or worker for Vultr. No unmeasured performance claims. [1]

Economic inclusion is a secondary discussion of service access, not a demonstrated fintech outcome. Healthcare, FDM and Solana are outside the build. Avoid extending the product merely to collect categories.

PROJECT PLAN / 10

Limits, demo and source key

Keep the evidence and the unfinished questions visible

Current limitations and remaining investigations

F1: corridor choice and stop-level baseline observations remain open. **F2:** runtime targets are unmeasured. **F4:** demo thresholds and ranking priority need values. **F5:** package effects, costs and full-cycle fleet assumptions need review. **F6:** live Papiers access is untested. **F7:** named owners and authoritative times need confirmation. **F3** is addressed by adding journey-time metrics; their implementation still needs checking.

A verified simulator follows its rules; that alone does not validate a real TTC forecast. Fixed transit share, uncertain demand, one direction, a short segment and simplified operations limit conclusions. Repeated experiments cannot repair wrong inputs. The agent discovers patterns and mechanisms within this model; it cannot establish unseen real-world causes, planning permissions or network-wide impacts.

Four-minute demonstration

0:00-0:40: corridor, developments and the move-in decision. **0:40-1:30:** growth, package tradeoffs and a downstream bottleneck. **1:30-2:50:** actual Papiers context, completed experiment and evaluation; a human correction changes the agent's next step. **2:50-3:30:** inspect the next configuration and phase/budget constraints. **3:30-4:00:** export the brief and explain uncertainty. Use identified saved runs if needed; distinguish them from new live computation.

Sources and input material

Public pages checked October 16, 2026. Links establish source availability and described capabilities; downloads, calibration and integration access are separate build gates. The detailed Papiers requirements come from the user-supplied challenge text.

[1] Hack the Valley 11 overview and rules | Rules

[2] Papiers Autoresearch Challenge (user-supplied full brief); product

[3] City of Toronto: Development Pipeline

[4] Toronto Open Data: TTC Routes and Schedules

[5] TTC: Transit Planning

[6] OpenStreetMap attribution and licence

[7] Toronto Open Data: 3D Massing

[8] TTC: RapidTO Surface Transit Network Plan

[9] Google: Gemini structured outputs

[10] CityCast: public product description

[11] UrbanSim: Scenario Modeling

[12] Replan: Transit and SIM

Also used: the supplied FutureProof concept/build brief and Chuds Brainstorming Session (1).pdf. Papiers is part of this plan's core workflow; signal priority is outside the build.